## Common Data Preprocessing Techniques

Data preprocessing is a task for cleaning and transforming raw data so it can be used for data analysis (MathWorks, n.d.). Some common techniques in data preprocessing include data cleaning, which fixes errors such as duplicate records and spelling mistakes (Bonnefoy, 2024). Another is handling missing values, which is done using a method called *imputation*, which reassigns an empty value with an estimated value. *Interpolation* is another technique that uses nearby data points to estimate a missing value (MathWorks, n.d.; Bonnefoy, 2024). There is also *normalization*, which changes the scale of numbers so they are easier to compare. For example, *min-max scaling* puts all numbers between a specific range, such as 0 and 1 (Mathworks, n.d). Finally, structural operations help organize data by grouping similar items or binning them into categories (MathWorks, n.d.).

## The Concept of Data Blending

Data Blending is the process of taking data from different places and putting it together into one clear structure (Bonnefoy, 2024). It is quite different from just moving data around since it actually harmonizes the information to make sense together. This is often done by joining different tables using a shared "key," like customer ID (MathWorks, n.d.). This concept is necessary because important information is usually scattered across many different systems, like databases, spreadsheets, and cloud APIs (Bonnefoy, 2024). Blending creates a unified view that lets people look at the whole picture instead of just the small pieces.

## The Importance of Blending for Informed Decisions

Blending multiple datasets is important because it helps find patterns and connections that are hidden in single sources (Bonnefoy, 2024). When data stays in separate "silos," it is hard to see how one thing affects another (Bonnefoy, 2024). Blending data can give companies some advantages in overseeing their operations and customer feedback. This leads to having more informed decisions being made based on having a complete set of facts rather than guesses. The risk of making mistakes gets reduced as we don't rely on incomplete information, helping businesses stay more competitive.

## Applying Techniques in Academics and Personal Data

In my own data work, I use techniques to make sure my results are as accurate as they should be with no mistakes. Preprocessing is a necessity for AI workloads. Standardization being applied helps models become more accurate and avoid being confused by noise in raw datasets. These methods ensure AI can learn effectively from the data being provided. Personally, I apply these same methods when I edit photos taken from events. I use *deduplication* to remove any extra copies of images when photoshooting in burst, filtering out only the good pictures. I also *blend* data from using editing software, like making color corrections and adding filtering layers, applying changes to RAW image file formats. Applying these techniques ensures that in each process, there is consistency in making changes, which in turn is meant to bring out the best final version of a product.

## References


MathWorks. (n.d.). Data Preprocessing.
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Bonnefoy. (2024). The Definitive Guide to Data Integration: Unlock the power of data integration to efficiently manage, transform, and analyze data. Packt Publishing.
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